Bachelor of Computer Application (BCA)

Veer Kunwar Singh University, Ara

Semester-wise Subjects, Topics, Full Marks and Pass Marks

Semester	Theory Subjects
I	Fundamentals of Computer; PC Packages; Programming Logic & Design Techniques; Programming in C; Communicative English
II	Digital Computer Organisation; Advanced Programming in C; Fundamental Data Structure; Application Programming in FoxPro; Financial Accounting
III	Information Technology Trend; GUI Programming in Visual Basic; Database Management System; Computer Networking & LAN; Management Skills
IV	Operating Systems; OOPS Programming in C++; Internet & E-Commerce; Linux Operating Systems; System Analysis & Data Processing
V	Component Architecture and Programming (COM, DCOM); Multimedia Tools & Applications; Programming in JAVA; Oracle RDBMS; Computer Centre Management
VI	Major Project

SEMESTER I

Subjects, Full Marks and Pass Marks

Theory Papers (Group A)

Paper Code	Subject	Full Marks	Pass Marks
1BCA1	Fundamentals of Computer	100	40
1BCA2	PC Packages (Windows 98/XP, Word, Excel)	100	40
1BCA3	Programming Logic & Design Techniques	100	40
1BCA4	Programming in C	100	40
1BCA5	Communicative English	100	40
Aggregate Marks		**500**	**225**

Practical and Sessional Papers (Group B)

Paper Code	Subject	Full Marks	Pass Marks
1BCA6	Internal Assessment & Term Work	100	45
1BCA7	Computer Lab - I	100	45
1BCA8	Professional Personality Skills	50	23
Aggregate Marks		**250**	**113**

Topics Studied in Each Subject

1BCA1 - Fundamentals of Computer **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction to Computers: Data; Information; Importance of Data and Information; Information Society and Information Economy; Computing Technology; Definition of Modern Day Computers; History of Computing (Napier's bones, Pascal, Difference Engine etc.); Common Applications of Computers; Definition of Information System and Information Technology; Multimedia

Unit 2 - Classification of Computers: Digital vs Analog Computer; Generation of Computers; General Purpose vs Special Purpose Computers; Micro-Mini-Mainframe-Super Computer; Single User-Multiuser System

Unit 3 - Data Representation: Number System - Decimal Number System; 9's and 10's Complements; Binary Number System; Binary Arithmetic, 1's and 2's Complements; Decimal to Binary Conversion; Binary to Decimal Conversion; Octal Numbers; Hexadecimal Numbers; Binary Coded Decimal System; Alphanumeric Code System (ASCII, EBCDIC); Bit; Byte

Unit 4 - Overview of Computer Architecture and Organization: Definition of Computer Architecture; Definition of Computer Organization; Structural Components of Computer System (CPU, Main Memory, I/O, System Interconnection); Structural Components of CPU (CU, ALU, Registers, CPU Interconnection); Basic Block Diagram of Computer; Basic Functions of Computers (Data Processing, Data Storage, Data Movement, Control); Major Components of Computer System (Hardware, Software); Application Software; System Software; Compiler, Interpreter and Assembler; Output Devices - Monitor, Printer (Impact, Non-Impact and Line, Dot Matrix, Laser, Drum and Chain Printers etc.); Memory Organization - registers, cache, primary and secondary memory; Memory associated concepts - Tracks, Sectors, Seek time, Latency time, Access time; Static RAM; Dynamic RAM; Input-Output Organization - Programmed I/O, Interrupt driven I/O, DMA; Input Devices (Keyboard, Mouse, Joystick, Scanner, MICR, DMR etc.)

Unit 5 - Overview of File System in Computers: Files; Directories; Types of Files; Internal and External Commands; Concepts of Path (Absolute and Relative path); File Management Commands of some common Operating Systems (Copy, Type, Xcopy, Edit, ls, cd, md, cat, vi etc.)

1BCA2 - PC Packages (Windows 98/XP, Word, Excel) **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction to PC Application Package: Definition and Example - Word Processing; Spreadsheet; DBMS; Presentation; CAD/CAM; Accounting

Unit 2 - Graphical User Interface: Operating System - an interface; Command Line User Interface; Graphical User Interface; GUI Components - Windows, Desktop, Icons, Types of Icons, Dialog Box, Menu, Command Button, Textbox, List Box, Radio/Option Button, Check Box, Spinner, Scroll Bar, Combo Box

Unit 3 - Windows Operating System: Different Versions/Types of Windows - Windows 3.1, 95, 98, Me, NT, 2000, XP; Basic Mouse Techniques - Click, Double Click, Point, Drag, Drag-and-Drop; Basic Keyboard Techniques - closing window/application, moving between open windows, selecting an application; Tasking help; Menu and its Convention - dimmed command, ellipsis, check mark, shortcut key to right of a command, triangle to the right of a command; Control Menu and its options - Restore, Move, Size, Minimize, Maximize, Close, Switch to; Navigating in a dialog box; Working with File Manager/Windows Explorer; Creating files; Directories/folder; Shortcut; Searching for files and directories; Moving; Copying; Renaming files and directories; Associating files with applications; Formatting floppy disk and hard disk; Scandisk; Disk Defragmentation; Setting Desktop environments; Printing; Applets of Windows - Notepad, Paint, Wordpad, Calculator, Character Map, Media Player; Users

Unit 4 - Document Creation and Word Processing: Introduction to Word Processing; Opening and closing MS-Word; Menu Bar; Tool Bar (Tool Buttons); Opening; Closing; Saving documents; Page setup; Display/Hiding of paragraph marks and inter-word spaces; Formatting - Font, Size, Style, Color, Alignments, Indentation, Bullets and Numbering, Tab and tab setting, Changing case; Handling Multiple documents - opening and closing of multiple documents, cut, copy and paste across the documents; Saving the clipboard; Moving around a document; Document template; Table Manipulation; Mail Merge; Macro; Printing

Unit 5 - Spreadsheet: Opening Spreadsheet (Excel); Menu bar; Tool bar (Tool buttons); Context Sensitive help; Introduction to common applications associated with spreadsheet - calculation of taxes, sales analysis, financial A/c, product scheduling, sale invoicing, statistical analysis and budgeting; Concepts - Workbook, Worksheet, Cell, Cell Pointer, Cell address, Cell range; Moving around worksheet; Entering Data - Text, Numbers, Date, Formula; Referencing - Relative, Absolute, Mixed; Editing; Specifying a Range; Formatting; Data fill; Sorting and Filtering; Form; Insertion of rows, columns, sheets; Changing width of rows and columns; Chart; Functions - Mathematical, Logical, Statistical, String, Financial, Database, Date and Time; Macro; Analysis tools - Goal seek, Solver, Scenario; Printing; Merging Cells

1BCA3 - Programming Logic & Design Techniques **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction: Origin and definition of the term 'Algorithm'; Euclid's algorithm; Features of algorithm - finiteness, definiteness, input, output, effectiveness

Unit 2 - Mathematical Preliminaries: Mathematical Induction; Numbers; Powers and Logarithms; Sum and Products; Integer Functions and Elementary Number Theory; Permutations and Factorials; Binomial Coefficients; Harmonic Numbers; Fibonacci Numbers; Matrices

Unit 3 - Algorithm Specification & Performance Analysis: Pseudocode; Flowchart; Performance analysis - space complexity, time complexity; Asymptotic Notation - Big 'Oh', Omega, Little Omega

Unit 4 - Programming Concepts: Programming Elements - Variable, Constant, Data Type, Array, Expression, Statements; Control Structure - Sequence, Selection, Iteration; Subroutines and Functions

Unit 5 - Software Development Methodology: Conventional Programming Technique; Structural Programming; Recursive Programming; Object Oriented Programming

1BCA4 - Programming in C **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction to C Language: (a) History; Program development - coding, compilation, linking, running; (b) Important features of C - structured programming, high/middle/low level, case sensitiveness, character set, block structure; (c) Different versions of C compilers

Unit 2 - C Vocabulary: Keywords; Operators - Arithmetic, Logical etc.; Operators precedence and associativity; Library Functions and their header files

Unit 3 - Data Types: Kinds, Range and Size; Memory variables; Constants; Basic data type modifiers

Unit 4 - Control Structures: if, switch, for, while, do-while

Unit 5 - Array: 1D, 2D, 3D

Unit 6 - Functions: Declaration, Definition, Calling, Making a library Function

Unit 7 - Storage Class: Storage class modifiers

1BCA5 - Communicative English **Full Marks: 100 | Pass Marks: 40**

Objective: To improve proficiency in English by developing skills in reading, writing, listening, speaking and composition. Kinds of composition: paragraphs, expository composition, narrative composition, descriptive composition, notes, reports and summaries

Writing Skills: 1. Business Letters (Correspondence); 2. Interpretation of various charts and graphs; 3. Vocabulary Building; 4. Paragraph Writing / Expansion of ideas / Comprehension; 5. Reportage

Speaking Skills: 6. Basic Sentence Patterns; 7. Narrating Stories with the help of hints; 8. Describing some incidents (hints minimised); 9. Conversation, Debating, Group Discussion; 10. Extempore Topics for Speech

SEMESTER II

Subjects, Full Marks and Pass Marks

Theory Papers (Group A)

Paper Code	Subject	Full Marks	Pass Marks
2BCA1	Digital Computer Organisation	100	40
2BCA2	Advanced Programming in C	100	40
2BCA3	Fundamental Data Structure	100	40
2BCA4	Application Programming in FoxPro	100	40
2BCA5	Financial Accounting	100	40
Aggregate Marks		**500**	**225**

Practical and Sessional Papers (Group B)

Paper Code	Subject	Full Marks	Pass Marks
2BCA6	Internal Assessment & Term Work	100	45
2BCA7	Computer Lab - II	100	45
2BCA8	Summer Assignments	100	45
2BCA9	Professional Personality Skill	50	23
Aggregate Marks		**350**	**158**

Topics Studied in Each Subject

2BCA1 - Digital Computer Organisation **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction to Digital Computers: Digital Computer vs Analog Computer; Characteristics of Digital Computers

Unit 2 - Binary Numbers, Codes and Arithmetic: Numbering system conversion; Binary addition and subtraction; Complement representation of numbers; Addition/Subtraction of numbers in one's complement notation; Addition/Subtraction of numbers in two's complement notation; Binary Multiplication and Division; Binary Coded Decimal Numbers; Floating point Arithmetic

Unit 3 - Digital Systems Algebra: Postulates of Boolean Algebra; Basic theorems of Boolean Algebra; Boolean functions and truth tables; Different forms of Boolean functions; Logic gates; Simplifying Boolean functions; Karnaugh map

Unit 4 - Combinational Switching Circuit: Combinational circuit design procedure; Binary operators and logic gates; Integrated circuits and NAND-NOR gates; Realization of Boolean expression with NAND gates; Some common combinational circuits used in digital systems

Unit 5 - Flip-Flops, Registers and Counters: Basic sequential circuit and types; Sequential circuits - flip-flops, counters, counter decoders, controlled counters, shift registers, push-down stack, transfer of information between registers; Single shot trigger

Unit 6 - Digital Memory Systems: Memory parameters; Characteristics of magnetic cores; Semiconductor; Random access memory; Read only memories; Content addressed memories

Unit 7 - Basic Machine Organization: Storage organization; Instruction and data representation; CPU organization; I/O organization; Basic instruction set; Base registers; Instruction formats

2BCA2 - Advanced Programming in C **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Abstract Data Types: Structure and union

Unit 2 - Pointers: Declaring pointer; Assigning address to a pointer; Size of pointer; Types of pointers; Pointers and array; String manipulation; Array of pointers

Unit 3 - Functions Revisited: Function call - recursion; Call by value and call by reference; Variable number of arguments; Command line arguments

Unit 4 - File Handling: Opening, closing and processing files using library functions

Unit 5 - Bitwise Operations: Bitwise operators >>, <<, !, &, ~ and their applications

2BCA3 - Fundamental Data Structure **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction to Data Structure: Array - a data structure (concept); Linear and non-linear data structure; Memory allocation technique - static and dynamic; Deallocation of memory

Unit 2 - List: Sequential list; Linked list; Single, double and circular linked list; Insertion - beginning, middle, end; Modification; Deletion; Searching; Traversing and merging linked list; Representing polynomial equations using linked list

Unit 3 - Stack: Sequential, linked, push, pop, display; Infix, postfix and prefix notation of arithmetic expression using stack

Unit 4 - Queue: Sequential, linked; Insertion, deletion and display

Unit 5 - Tree: Sequential and linked representation; Binary search tree - insertion; Traversing (in order, pre order, post order) recursively and non-recursively; Deletion in B.S.T.; Concept - AVL tree, Threaded binary tree, B-tree; Representing arithmetic expression using binary tree

2BCA4 - Application Programming in FoxPro **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction: FoxPro - an introduction; Concept of DBMS and RDBMS; Files, Tables, Records, Fields

Unit 2 - FoxPro Database Management Utilities: FoxPro working environments; Creating, viewing, editing and deleting table structure and its contents; Sorting and indexing database files; Reports and labels

Unit 3 - Memory Variable & Functions: Creating and using memory variables and arrays; Saving and restoring memory variables; Setting environment commands; Time and date functions; Mathematical commands and functions; Data type conversion functions; Macros; Character functions

Unit 4 - Programming with FoxPro: Creating a program file; Performing input/output; Control structure - IF, SCAN, FOR, DO WHILE, TEXT; Modular approach - functions and procedures; Programming with multiple database files; Programming with multiple program files in a project; Creating program interface - boxes, windows, menus and pop ups; Key trapping and manipulation; File manipulation functions

Unit 5 - Exceptional Conditions: Different error conditions; Locating errors; Common error messages; Error debugging techniques

2BCA5 - Financial Accounting **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction to Accounting: What is accounting?; Difference between accounting and book keeping; Advantages of accounting; System of financial accounting; Classification of accounts; Rules of debit and credit; Manual vs computerized accounting

Unit 2 - Voucher: Receipt and payment voucher; Journal voucher; Sale and purchase voucher; Debit and credit note; Reverse journal and memo voucher

Unit 3 - Accounting Group: Groups and sub groups; Primary and secondary cost centres

Unit 4 - Account Books: Cash/Bank book; Journal; General ledger; Memorandum register; Sales register; Purchase register; Day book

Unit 5 - Accounting Information & Reports: Trial balance; Profit and loss; Balance sheet; Ratio analysis; Cash flow; Fund flow; Statements of accounts; Negative ledger; Overdue receivable; Overdue payables; Adjustment

SEMESTER III

Subjects, Full Marks and Pass Marks

Theory Papers (Group A)

Paper Code	Subject	Full Marks	Pass Marks
3BCA1	Information Technology Trend	100	40
3BCA2	GUI Programming in Visual Basic	100	40
3BCA3	Database Management System	100	40
3BCA4	Computer Networking & LAN	100	40
3BCA5	Management Skills	100	40
Aggregate Marks		**500**	**225**

Practical and Sessional Papers (Group B)

Paper Code	Subject	Full Marks	Pass Marks
3BCA6	Internal Assessment & Term Work	100	45
3BCA7	Computer Lab - III	100	45
3BCA8	Professional Personality Skills	50	23
Aggregate Marks		**250**	**113**

Topics Studied in Each Subject

3BCA1 - Information Technology Trend **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Information System: Hardware: The computer structure, architecture; The memory system; I/O technology; Selection of computer system; Communicating system hardware

Unit 2 - Information System: Software: The system software; File system; Directory structure; Networking software; Network security; Graphical User Interfaces; Application software

Unit 3 - Software Engineering: Definition and taxonomy; Software development paradigms; Requirement analysis and specification; Design strategies; Design tools and diagramming aids; Design concepts; The relationship between analysis and design and how to make the transition; Various design methodologies

Unit 4 - Operating System: An overview of operating system; Distributed operating system; Processor management; I/O file management

Unit 5 - Networking Concept: Data communication concept; Classification - serial, parallel, simplex, half-duplex, duplex; Communication media - wire, microwave; Internet; Intranet (basic concepts); Hardware/software components required for LAN; Topologies; Protocols (introduction); OSI layers (introduction)

3BCA2 - GUI Programming in Visual Basic **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction: Visual Basic - a Rapid Application Development tool; Integrated Development Environment of Visual Basic; Concept of object oriented and event driven programming; History of Visual Basic

Unit 2 - Visual Basic Controls: Concept of control/ActiveX control; Label, Textbox, Command Button, Option Button, Check Box, List Box, Combo Box, Slider Control, Image, Picture, Scroll Bar, Clock, File, Directory, Drive, Timer, Tab Strip, Flexgrid, Dialog Control; Important properties, methods and events; Menu Editor; Tool Bar; Designing interface using controls; Menu and Tool Bar

Unit 3 - Declaration and Using Memory Space: Variables; Data types; Scope of variables; Local and global variables; Static variables; Constants; Array; Dynamic array; Control arrays

Unit 4 - Control Structure: Sequence, selection and iteration structure; Selection and iteration control statements

Unit 5 - Procedures & Functions: Creating and working with event driven module; Creating and using sub module and function module; Module; Call by value, call by reference; Recursion; Library functions - string, date and time, mathematical and financial; Type conversion

Unit 6 - Exception Handling: Introduction to exceptional situations - design time, compile time and run time errors; Error handling techniques

Unit 7 - File and Database Handling: Different file organization; Files related controls, objects and properties; Working with sequential access file; Working with random access file; Working with DBMS in VB; Report generation in VB

Unit 8 - Object Oriented Programming: The concept of abstraction and encapsulation - class, object; Class module vs standard module; Creating and using classes; Implementing polymorphism in Visual Basic; Introduction to creating ActiveX control

Unit 9 - VB Project Deployment: Making an exe file; Package and deployment wizard

3BCA3 - Database Management System **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction: Introduction to Database Management System; File systems and associated problems; Advantages and disadvantages of DBMS; Component/structure of DBMS; Different data models - introduction to relational, hierarchical and network model

Unit 2 - Data Models: Concept of entities, attributes, association and relationship; Entity Relationship model; Drawing E-R diagram; Data abstraction - generalization, specialization, aggregation; ANSI-SPARC model

Unit 3 - Relational Data Model, Implementation & Manipulation: Concepts of attributes and domains; Tuples; Keys; Introduction to Codd's rules; Introduction to relational algebra and relational calculus; SQL - DDL, DML, DCL; Introduction to QUEL; Introduction to RQBE

Unit 4 - Normalisation: Relational database design issues; Anomalies in database; Introduction to functional dependency; Normalization - 1NF, 2NF, 3NF, BCNF; Normalization through synthesis

Unit 5 - Database Security: Different security policies - authorization, identification, authentication, encryption/decryption, integrity rules and constraints, auditing; Introduction to database recovery mechanism

Unit 6 - Future Trends in DBMS: Concept of distributed database system; Knowledge based database system; Expert database system; Object database system

3BCA4 - Computer Networking & LAN **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction to Computer Network: Computer network and uses; Network goals; Applications of networks

Unit 2 - Network Structure: Hosts; Communication subnet; IMPs; Point-to-point channels and different topologies of point-to-point channels; Broadcast channels

Unit 3 - Network Architecture: Protocol hierarchies; Peer processes

Unit 4 - OSI Reference Model: Different layers of OSI model; Connection oriented and connectionless services

Unit 5 - Transmission Media: Magnetic media; Twisted pair; Baseband coaxial cable; Fibre optics; Communication satellites

Unit 6 - Transmission & Switching: FDM and TDM; Circuit switching; Packet switching; Hybrid switching; ISDN

Unit 7 - Broadcast Networks & their Protocols: CSMA/CD; CSMA/CA; Ethernet (802.3); Types of Ethernet cables; Token bus (802.4); Token ring (802.5); WAN repeaters; Bridges; Routers; Gateways; FDDI backbones; TCP/IP

3BCA5 - Management Skills **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction to Management: Definition; Nature and scope; Concepts of management; Administration and organisation; Management process and levels of management

Unit 2 - Evolution of Management Thought: Contribution of F. W. Taylor, Henri Fayol, Peter Drucker, Henry Mintzberg, Tom Peters etc.

Unit 3 - Planning: Scope; Importance; Limitations; Procedure of planning; Types of plans; Mission, objective, policy, strategy, procedure and rule; Types of planning - strategic and tactical

Unit 4 - Organising: Procedure and principles; Span of control; Concepts of authority and responsibilities; Delegation and decentralization; Line, staff and functional authority; Types of organisation structure

Unit 5 - Leading: Leadership theories; Traits and styles; Motivation - theories of Maslow, McGregor, Herzberg, McClelland, Lawler-Porter model

Unit 6 - Controlling: Process and types of control

SEMESTER IV

Subjects, Full Marks and Pass Marks

Theory Papers (Group A)

Paper Code	Subject	Full Marks	Pass Marks
4BCA1	Operating Systems	100	40
4BCA2	OOPS Programming in C++	100	40
4BCA3	Internet & E-Commerce	100	40
4BCA4	Linux Operating Systems	100	40
4BCA5	System Analysis & Data Processing	100	40
Aggregate Marks		**500**	**225**

Practical and Sessional Papers (Group B)

Paper Code	Subject	Full Marks	Pass Marks
4BCA6	Internal Assessment & Term Work	100	45
4BCA7	Computer Lab - IV	100	45
4BCA8	Minor Summer Assignment	100	45
4BCA9	Professional Personality Skills	50	23
Aggregate Marks		**350**	**158**

Topics Studied in Each Subject

4BCA1 - Operating Systems **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction to Operating System: What is an operating system; Functions of operating systems; Interaction of operating system with hardware and user programmes; Types of operating systems (batch, multiprogramming, network, distributed, time sharing); Structure of an operating system

Unit 2 - Process Management: Definition of process; Process states; Process state transition; The process control block; Operation on process (suspend and resume); Mutual exclusion; Critical section; Dekker's algorithm; Semaphores; Deadlock; Indefinite postponement; Necessary conditions for deadlock; Deadlock prevention, avoidance, detection and recovery

Unit 3 - Storage Management: Storage organisation, management, hierarchy, management strategies; Contiguous and non-contiguous storage allocation; Single user contiguous storage allocation; Variable partition multiprogramming; Virtual storage; Paging; Segmentation; Combined paging and segmentation system; Page replacement strategies; Demand paging; Page fault frequency page replacement

Unit 4 - Processor Management: Scheduling levels; Scheduling criteria; Pre-emptive vs non-preemptive scheduling; Interval timer; Priorities; FIFO scheduling; RR scheduling; SJF scheduling; SRT scheduling

Unit 5 - File Management: File concept; Directories; Directory structure in DOS; Moving, renaming, copying, deleting and undeleting files under DOS; Disk organisation; Disk scheduling

Unit 6 - Device Management: Control of various devices; Device drivers; Interrupt driven or poll driven data transfer; Need of software and hardware protocols

4BCA2 - OOPS Programming in C++ **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction to Object Oriented Concepts: Inheritance; Class; Objects; Polymorphism, overloading; Dynamic binding; Advantages of OOP; Object oriented analysis and design

Unit 2 - Fundamentals of C++ Programming: History of C++; Structure of a C++ program; Declaration of class and object - definition and declaration of members; Objects as data types; Objects as function arguments; Structures and classes; Data types - basic and user defined; Variables and constants - dynamic initialization of variables, reference variables, enumerated variable; Operators and expressions; Coding control constructs - sequence, selection, iteration; Arrays and strings; Input/output mechanisms: cin, cout; Concepts of functions - declaration, calling, definition; Scope rules of members of class; Data and type conversion - between basic types, between objects and basic types, between objects of different classes; Limitations of type conversion

Unit 3 - Constructors and Destructors: Basic constructors; Parameterized constructors; Constructors with default arguments; Dynamic initialization of objects; Copy constructors; Destructors; Limitations on constructors and destructors

Unit 4 - Overloading: Function overloading; Operator overloading - unary, binary

Unit 5 - Inheritance: Derived classes and base classes; Access specifiers; Derived class constructors; Member function overriding; Types of inheritance; Public and private; Multiple inheritance

Unit 6 - Pointers and Virtual Functions: Pointer concepts; Pointers to objects; Memory management using new and delete operators; Virtual functions and polymorphism; Friend functions; Static functions; Macros and inline function

Unit 7 - Streams: Stream classes - hierarchy; Header files; Stream manipulators; String stream; Character streams; Object I/O; File streams; Disk I/O with member function

Unit 8 - Templates: Function templates; Class templates

Unit 9 - Exception Handling: Try block; Catch handler; Throw statement; Exception specification

4BCA3 - Internet & E-Commerce **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Internet Concept: What is Internet; How it is different from Intranet; Groupware; Features of Internet - cheaper, flexible, versatile; Data transmission in Intranet; Platform for Internet; Network topologies; Servers (NT/Novell/UNIX/Linux); Client workstation

Unit 2 - Internet Implementation: Hardware, software and protocols for Internet; IP network; HTTP server; SMTP, POP3, IMAP4, LDAP, document server; Implementing Intranet using Novell and Windows

Unit 3 - Electronic Commerce and Banking: Requirements; Standards; Case studies

Unit 4 - Electronic Commerce and Retailing: Requirements; Standards; Case studies

Unit 5 - Electronic Commerce and Online Publishing: Requirements; Standards; Case studies

4BCA4 - Linux Operating Systems **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction to System Software: System software and types of software

Unit 2 - Introduction to Assembler: Types of translator; Assembler; Implementation; Macro and macroprocessor; Loaders

Unit 3 - Introduction to Compiler: Approaches to compiler development; Compiler design phases; Software tools lex, yacc

Unit 4 - Introduction to OS: Operating system structure - layered structure approach, kernel approach, virtual machine, client-server model

Unit 5 - Basics of Linux: History of Linux; Features of Linux; System software layers

Unit 6 - File System & Concepts of Block in Linux: File system; Files and directories commands; Different types of files and directory permission; Other useful commands; Disk related commands; Boot block; Super block; I/O redirection; Piping; Tree command; Filters

Unit 7 - Communication in Linux: Process in Linux; Background and foreground processes; Advantages and disadvantages of background process; Running process in background; Different process related commands

Unit 8 - Communication in Linux: Communication in Linux; Different communication related commands - write, wall etc.

Unit 9 - Shell Programming: vi editor; Shell variables; Keywords; Positional parameters; Passing command line arguments; Control instructions etc.

4BCA5 - System Analysis & Data Processing **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Software Life Cycle: Models - Waterfall, Spiral, Prototyping, Fourth generation techniques, SW process; Software Requirements Specification (SRS); Fact-finding techniques; Characteristics of a good SRS - unambiguous, complete, verifiable, consistent, modifiable, traceable and usable during the operation and maintenance phase; Prototype outline for SRS

Unit 2 - SW Inspection: Communication skills for the system analyst; Review/inspection procedure; Document; Composition of the inspection team; Checklist; Reading by the inspectors; Recording of the defects and action recommended; Practice of inspecting small requirement specifications for good characteristics

Unit 3 - System Analysis: SA tools and techniques; DFD; Entity Relationship diagrams; Project dictionary

Unit 4 - SW Design: System design tools and techniques; Prototyping; Structured programming

Unit 5 - User Interface Design: Elements of good design; Design issues; Features of a modern GUI; Menus; Scrolling; Windows; Icons; Panels; Error messages etc.

SEMESTER V

Subjects, Full Marks and Pass Marks

Theory Papers (Group A)

Paper Code	Subject	Full Marks	Pass Marks
5BCA1	Component Architecture and Programming (COM, DCOM)	100	40
5BCA2	Multimedia Tools & Applications	100	40
5BCA3	Programming in JAVA	100	40
5BCA4	Oracle RDBMS	100	40
5BCA5	Computer Centre Management	100	40
Aggregate Marks		**500**	**225**

Practical and Sessional Papers (Group B)

Paper Code	Subject	Full Marks	Pass Marks
5BCA6	Internal Assessment & Term Work	100	45
5BCA7	Computer Lab - V	100	45
5BCA8	Professional Personality Skills	50	23
Aggregate Marks		**250**	**113**

Topics Studied in Each Subject

5BCA1 - Component Architecture and Programming (COM, DCOM) Full Marks: 100 | Pass Marks: 40

Unit 1 - COM & DCOM Basics: COM/DCOM issues; Components; Persistence; Sharing and scalability

Unit 2 - Building Distributed, Scalable Applications with COM, Part I: Transactions and databases; Multi-tier architecture; Security

Unit 3 - Building Distributed, Scalable Applications with COM, Part II: Clustering; Message queuing

5BCA2 - Multimedia Tools & Applications Full Marks: 100 | Pass Marks: 40

Unit 1 - Introduction: Goal, objective and applications; Multimedia building blocks; Multimedia hardware, platform and software tools; Authoring tool

Unit 2 - Multimedia System: File formats (BMP, GIF, PCX etc.); Communication system; Multimedia database

Unit 3 - Video Technology: Animation basics; Digital imaging (still and moving); Video capture; Video processing; Video recovery techniques; AVO and AVI file formats; Conferencing techniques; Video streaming; Video technology - motion of synchronization

Unit 4 - Video Compression: LZW; DCT coding; JPEG; MPEG; Hypertext and hypermedia; SGML; Document architecture; ODA; MHEG

Unit 5 - Hardware Peripherals: Connection; Memory and storage devices; Input and output devices; Communication devices

Unit 6 - Multimedia & Internet: Design WWW

Unit 7 - Multimedia on Windows: Multimedia function calls; Windows support for sound, animation, movies, music and MIDI controls

5BCA3 - Programming in JAVA **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Applet as Java Applications: Applet specific methods and related HTML references; Creating an applet; Displaying it using web browser with an applet viewer; Advantages and disadvantages of applet vs applications

Unit 2 - Abstract Window Toolkit and Event Handling in Java: Components used in AWT; Basics of event handling and individual events; AWT package

Unit 3 - Multithreading: Introduction; Thread classes and its methods; Thread priorities; Thread synchronization

Unit 4 - Swing Libraries: The model-view-controller design pattern; Different layouts, menus, dialog boxes, text input

Unit 5 - Input / Output using Streams: Introduction; Files and streams; Reading data from sequential access file; Updating sequential access file; Random access file; Class file; BufferedReader and BufferedWriter classes

Unit 6 - Introduction to Networking with Java: Introduction; The URL class; The socket and server-socket classes; Connecting to the server; A simple chat program

Unit 7 - JDBC: The Java database connectivity standard; 4 different types of drivers and their advantages; JDBC-ODBC connectivity with MS-Access database; Three-tier architecture

Unit 8 - Java Beans & EJB: Why Beans; How Beans are used - the BDK; Beans properties and methods; Introduction to EJB

5BCA4 - Oracle RDBMS **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction to the Relational Model: An overview of the relational model; Data independence; ER diagrams; Relational algebra operations; Referential integrity and database integrity; SQL

Unit 2 - Normalisation and Query Processing: Functional dependency; Normalisation; Multivalued and join dependency; Query processing

Unit 3 - Recovery, Concurrency Management and Database Security: Reliability; Concurrency management; Concurrency control; Locking scheme; Database security, integrity and control; Practicals on SQL Server

5BCA5 - Computer Centre Management **Full Marks: 100 | Pass Marks: 40**

Unit 1 - Introduction: What is organization; Components of organization; Nature and variety of organization (in terms of objectives, structure etc.); Importance of organizational behaviour in work situation; Fundamental concept of OB; Different models of OB - autocratic, custodial, supportive, collegial

Unit 2 - Profile of an Entrepreneur: Perception; Attitudes; Commitment; Values; Creativity and other personality factors; Motives - primary, secondary (achievement, power, affiliation)

Unit 3 - Motivation Process: Intrapersonal conflict; Defence mechanisms; Study of selected theories of motivation - McGregor's Theory X and Y, Abraham Maslow's theory of need hierarchy, Herzberg's two factor theory, Victor Vroom's expectancy theory

Unit 4 - Group Dynamics: Formal and informal groups; Types of groups; Theories of group formation

Unit 5 - Organizational Design: Various organizational structures and their effects on human behaviour; Organizational climate; Organizational culture

Unit 6 - Leadership: Definition; Importance to the organization; Different styles of leadership

SEMESTER VI

Subjects, Full Marks and Pass Marks

Paper Code	Subject	Full Marks	Pass Marks
6BCA1	Major Project	500	225
6BCA2	Internal Assessment & Term Work	100	45
Aggregate Marks		**600**	**270**

What is Done in Semester VI

This semester has no theory papers. It consists only of the Major Project (6BCA1) and Internal Assessment & Term Work (6BCA2). Both are evaluated internally in the presence of at least two external examiners of the University.

6BCA1 - Major Project (Full Marks: 500 | Pass Marks: 225, i.e. 45%)

- The project may be done individually, or in a group in the case of a bigger software development project. In a group project, each student must be given responsibility for a distinct module, and the progress of individual modules must be independent of the others.
- Students take guidance from an internal guide and prepare a project report on 'Project Work' with at least 4 typed copies of the project synopsis. A separate file containing source code and listings is also submitted on floppy/CD.
- The college forwards one copy of the synopsis to each external panel member in advance of the project viva date.
- The project should be of such a nature that it is useful or relevant from the commercial/management angle.
- The project report is prepared in the format prescribed by the college, which specifies the contents and the method of presentation.
- The project should aim at developing software packages useful in automation of offices and organisational management, computerisation of billing systems, results, inventories etc.
- The project may be carried out in an industry, business establishment, educational organisation, consultancy etc. outside the college, with prior permission of the college. Students must report back to the college periodically to inform the internal guide about the development of their work.
- The project report is assessed by the internal guide, and the remark/grade is communicated by the Co-ordinator to the external examiner at the time of viva.
- On the scheduled date, students present 4 copies of the project record as a booklet along with floppy/CD. Marks are awarded on the basis of project presentation and demonstration before the departmental committee and the external examiners of the University.
- The viva-voce examination on the project report is conducted by a minimum of two external examiners at the end of the VIth semester.

6BCA2 - Internal Assessment & Term Work (Full Marks: 100 | Pass Marks: 45)

- Students appear for the Internal Assessment examination. Marks are awarded on the basis of class assessment, attendance, laboratory work report, and overall behaviour and discipline.
- Evaluation is done by a committee of teachers constituted by the Director / Principal / Head / Co-ordinator / Professor-in-Charge, and the marks are forwarded to the University.